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Data-Pooling Reinforcement Learning for Preventative Healthcare Intervention

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Abstract. Motivated by the emerging needs of tailored intervention for heterogeneous populations in preventative healthcare applications, we consider a multistage, dynamic decision-making problem in the online setting with unknown model parameters. To deal with the pervasive issue of small sample size in parameter estimation, we develop a novel data-pooling reinforcement learning (RL) algorithm that adaptively pools historical data, with three main innovations: (i) the weight of pooling is directly tied to the performance of the decision (measured by regret) as opposed to estimation accuracy in conventional methods; (ii) no parametric assumptions are needed between historical and current data; and (iii) requiring data-sharing only via aggregate statistics, as opposed to patient-level data. Our data-pooling algorithm framework unifies a variety of RL algorithms, and we establish a theoretical performance guarantee that quantifies the regret bound reduction from our adaptive pooling design compared with benchmarks. We substantiate the theoretical development with extensive empirical experiments via a case study in the context of post-discharge intervention to prevent unplanned readmissions, generating practical insights into healthcare management. Our algorithm enables organizations to leverage public data or published studies for patient management and supports policymakers in promoting aggregate data sharing to enhance population health outcomes.

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1. Introduction

Preventative interventions are important strategies to improve delivery of healthcare and population health. For instance, lifestyle interventions, including customized exercise programs and dietary changes, have emerged as effective strategies in promoting patient well-being (Anderson and Durstine 2019, Santos 2022). Phone calls or visits to enhance medical adherence have also demonstrated effectiveness in preventing infectious diseases and reducing avoidable hospital readmissions (Kini and Ho 2018, Vernon et al. 2019, Wilder et al. 2020). These interventions, characterized by their noninvasive nature and minimal associated risk, have been applied across diverse healthcare scenarios. Notable successes have been achieved in managing major public health priorities, such as diabetes (Newman et al. 2018), depression (Löwe et al. 2004), hypertension (Liao et al. 2020), and maternal health (Arpita et al. 2021). On the one hand, efficient intervention can improve individual health outcomes, and reduce expenditures and burdens on the healthcare systems. On the other hand, interventions are costly, for example, they require additional resources to staff healthcare workers to provide preventative interventions (Arpita et al. 2021). Therefore, it is imperative for decision makers in the healthcare and public health sectors to manage these tradeoffs. Dynamic intervention management has been a long-standing topic in the operations research fields (Tunc et al. 2014). In recent years, with the advancement of data analytics and integration of health information systems, there is a burgeoning interest in tailoring intervention strategies to different patients to best suit their individual needs and health condition changes (Ayer and Chen 2018). This research aims to address one critical challenge in data-driven intervention: the *small-sample issue*. Next,

we elaborate on the small-sample issue and its associated challenges, which are pervasive not only in many healthcare problems, but also in domains beyond healthcare—hence, of broad significance.

1.1. Background and Challenges

Markov Decision Process (MDP) is a common modeling tool to dynamically optimize preventative intervention planning. Examples include preventative screening for cancers (Ayer et al. 2016, Bertsimas et al. 2018), wearable devices for chronic disease management (Liao et al. 2020, Gautier et al. 2021), and post-discharge follow-ups with patients via phone calls or home visits to prevent hospital readmissions. These studies often assume a common model of the condition progression for the population. MDP-based personalization uses updated action or belief based on individual-specific observations yet the underlying transition matrix is still the same (Piri et al. 2022). Recent research has begun to do risk stratification based on regression or advanced machine learning models to capture the heterogeneity among different individuals in terms of the intervention effect and condition evolution; see Ayer and Chen (2018) for a comprehensive survey. However, these works typically follow the estimate-then-optimize (ETO) framework, where the MDP model parameters are first estimated and then used to determine the optimal policies under the assumption that these estimates are accurate. ETO and other offline methods can be particularly problematic when there is a potential mismatch between historical data and the characteristics of new (future) patients.

Consider a motivating example: A large health system opens a new hospital and needs to manage patient interventions there. While the system has extensive historical data from other locations, the patients at the new hospital (the "target patients") may differ significantly from historical populations. As a result, policies estimated purely from historical data may be suboptimal due to poor out-of-sample performance. This motivates the use of *online learning*, where policies are optimized while gradually learning model parameters from new target patient data, potentially leveraging biased historical data.[1] Specifically, we consider a reinforcement learning (RL) framework for multistage online decision-making. This framework is more suitable than static approaches (e.g., bandit learning) because current decisions can have long-term impacts on patient outcomes. However, our setting introduces three key challenges.

The first challenge arises from the limited data available from target patients. Standard RL algorithms that rely solely on target data require exploration of potentially suboptimal policies to account for estimation errors. When the target sample size is small, this leads to high variance in parameter estimates and requires more exploration than necessary. For instance, in our case study, this results in over 50% worse performance when only a few dozen target observations are available. The "small-data" issue is prevalent in personalized decision-making across various applications (Gupta 2022).

While regression-based methods attempt to address the first challenge by leveraging historical data through contextual learning, they rely on critical assumptions about the similarity between historical and target populations. These assumptions often fail in healthcare settings, where different locations may have vastly different social determinants and other unmeasured factors that affect outcomes (Tomkins et al. 2021). This poses a second challenge, impacting both offline algorithms (e.g., regressing on historical data and applying estimated policies to new patients) and regression-based transfer learning methods. As we will demonstrate analytically and numerically, simply merging historical and target data or relying solely on historical data can significantly degrade performance.

A third challenge is the need to share individual-level information during the learning process when multiple datasets are used. This is particularly difficult across different health organizations due to regulatory restrictions. It also limits researchers' ability to utilize public datasets or published studies, which often contain only aggregate-level data.

1.2. Overview and Contributions

In this paper, we develop a novel *data-pooling* RL method to address the three challenges in a holistic manner. Our method leverages historical data—using only aggregate statistics—to facilitate learning and decision-making in data-limited environments, with a provable performance guarantee. It efficiently combines potentially biased offline (historical) data with online (target) data. The design of our data-pooling method enables the use of historical data to reduce variance, as opposed to pure online algorithms, while properly accounting for bias, unlike pure offline estimates. In other words, it strikes a variance-bias trade-off and outperforms both pure online and offline algorithms. We highlight our contributions as follows.

I. **Model-free Design.** We formalize our learning framework in Section 3, which sets the structure of augmenting target data with historical data to address the first challenge. A salient feature of our framework is the *model-free* design, which addresses the second challenge brought by parametric assumptions in contextual methods. Moreover, it only requires aggregate statistics from historical data, thereby addressing the third challenge.

II. **Data-pooling for Decision**. The idea of using data from others to augment target data traces back to the empirical Bayesian literature (Efron and Hastie 2016), where the key is to establish the *proper* pooling mechanism. Our innovation lies in that the derivation of this pooling mechanism directly ties to improving

the decision quality, which sets it apart from conventional approaches focused on improving prediction accuracy. In Section 4, we formulate a "meta" PVI-λ algorithm in the tabular setting and link its regret performance (decision quality) through a set of confidence radii. Through this link, we design our pooling mechanism by setting the optimal weight λ^* to minimize these radii and establish the corresponding regret bound. In Section 5, we explicitly quantify the improvement of our PVI-λ^* algorithm over two benchmarks in a "small-data" limit regime. This innovative framework, linking data-pooling design with regret, holds substantial theoretical value and can be generalized to other RL problems.

III. **Practical Applicability.** In Section 6, we discuss extensions of our baseline pooling algorithm for practical applications. In Section 7, we demonstrate the empirical success of our algorithm via a case study on postdischarge intervention using real-world and synthetic data. We analyze the key drivers of its superior performance over benchmarks and summarize practical implications.

2. Literature Review

2.1. Tailored Intervention

Ayer and Chen (2018) provide a comprehensive survey on personalized interventions to contrast with traditional population-level intervention that typically uses a one-size-fits-all strategy; see references therein. Unlike most MDP-based research on preventative interventions, we focus on the joint learning of individual risk trajectories and decision-making. Our work also differs from (i) studies focused on prediction, such as predictive models for estimating readmission risk (Min et al. 2019), and (ii) papers utilizing the ETO framework. Skandari and Shechter (2021) study a related patient intervention problem. They assume that patient types are unknown but the dynamics for each type are given, making the primary task identifying patient types and adjusting treatment accordingly. This is modeled as a POMDP, where the patient type serves as the information state. In contrast, our framework does not assume predefined dynamics for known types; instead, both the dynamics and the optimal policy must be learned simultaneously. A related field is personalized clinical trial design (Chick et al. 2022), which primarily focuses on identifying the best treatment strategy. Here, the assigned strategy remains fixed for a patient, meaning individual state transitions are typically not considered. An exception is Bonifonte et al. (2022), who propose offline, population-based blood pressure management strategies to guide trial design, where state changes (blood pressure) are modeled via a Brownian motion.

2.2. RL in Healthcare Interventions

Bandit learning has been used in personalized healthcare interventions; see, for example, Zhu et al. (2018) and Keyvanshokooh et al. (2019). We focus on multistage sequential decision making in the online learning setting. The dynamic nature greatly complicates our problem. There is no simple heuristic to bypass the need for estimating risk evolution (Anonymous 2025). In multistage learning, two common regression-based approaches are clustering and contextual learning. Clustering methods first group patients based on features and treatment responses, then assign new patients to these clusters when prescribing policies (Utomo et al. 2019, Shi et al. 2021). Contextual learning methods use regression models to estimate value functions (Zhou et al. 2018, Liao et al. 2020). We analytically (section 1.3 of Anonymous 2025) and numerically (Section 7.4.4) demonstrate the limitations of these methods due to model misspecification. Our model-free algorithm offers built-in robustness to misspecification.

2.3. Data-Pooling and Transfer Learning

While the use of historical data traces back to empirical Bayesian literature (Efron and Hastie 2016), data pooling in decision-making is fundamentally different. Gupta and Kallus (2022) compare data pooling for estimation versus optimization in a stochastic offline setting, where all data are precollected and fixed throughout the optimization process—distinct from the online setting we consider. For online decision-making, data pooling has been used under the contextual bandit framework. Tomkins et al. (2021) study personalized treatment in mobile health applications and develop an adaptive pooling method that integrates empirical Bayesian principles with Thompson Sampling (TS) on Gaussian regression models. Bastani et al. (2022) propose a meta-learning algorithm for dynamic pricing, leveraging sequential TS experiments to improve learning across contexts. We stress that the RL setting has unique challenges so methods for the bandit setting do not directly extend. Among others, Miao et al. (2022) is the most closely related to ours, as they also adopt a model-free approach to handle data from different populations. Their method uses a semimyopic pricing policy combined with adaptive clustering to pool data across customer classes. In our case study, we provide a detailed comparison and show the advantages of our method.

Our work aligns with the growing literature on improving online learning algorithms using offline data. Bu et al. (2020) study a bandit setting where offline data shares the same distribution as the true system (analogous to $\Delta = 0$ in our setting) and show that it can significantly reduce regret. Gur and Momeni (2022) examine regret reduction when contextual data, which accurately predicts average rewards, arrives

randomly during online learning. In contrast, our setting involves offline data from a different population than the online data, introducing a difference gap Δ. This necessitates a careful balance between variance reduction and bias when pooling offline and online data. Our work also relates to transfer learning in RL, particularly when the source and target models share the same state-action space but differ in transition and reward parameters (Lazaric 2012). Most studies in this area focus on policy-transfer approaches, such as policy distillation (Parisotto et al. 2016, Rusu et al. 2016, Yin and Pan 2017) and policy reuse (Fernández and Veloso 2006, Barreto et al. 2016). Our method belongs to parameter-transfer, a meta-learning approach (Gupta et al. 2018, Xu et al. 2018), where the weight λ functions as a meta-parameter. Unlike standard meta-learning, which trains meta-parameters numerically, our algorithm derives them through theoretical analysis. In addition, while Parisotto et al. (2016) and Ferns et al. (2004) discuss convergence and performance bounds in transfer learning, to the best of our knowledge, there are no theoretical results on regret bounds in this setting. For comprehensive surveys, see Lazaric (2012) and Zhu et al. (2023).

2.4. Regret Analysis for RL

Our theoretical analysis is grounded in regret analysis for finite-horizon RL problems, a well-established area with a vast body of literature; see, for example, Auer and Ortner (2006), Jaksch et al. (2010), Wen (2014), Osband et al. (2016), Azar et al. (2017), and Russo (2019). While our regret bound framework builds on Russo (2019), our primary contribution lies in developing an effective data-pooling mechanism to enhance performance and rigorously characterizing its impact on regret bounds. This represents a significant and novel theoretical advancement.

3. Model Formulation

In this section, we specify our modeling and algorithm framework. We introduce the MDP model in Section 3.1 and the learning framework that uses historical data in Section 3.2.

3.1. MDP Model

We consider a finite-horizon MDP $\mathcal{M} = (\mathcal{H}, \mathcal{S}, \mathcal{A}, R, P)$ for each type of *target* patients, where $\mathcal{H}$, $\mathcal{S}$ $\mathcal{A}$ are the decision horizon, state space and action space, respectively. In particular, $\mathcal{H} = \{1, 2, \ldots, H\}$ and we assume that $\mathcal{S}$ and $\mathcal{A}$ are both finite sets. At the beginning of each epoch h, the state $s \in \mathcal{S}$ is observed first, then an action $a \in \mathcal{A}(s,h) \subset \mathcal{A}$ is taken, where $\mathcal{A}(s,h)$ is the feasible set of actions at epoch h when the patient is in state s. Once a is taken, the system evolves to state s' at epoch $h+1$ with probability $P(s'; h, s, a)$. To align with the learning literature, we consider maximizing reward as the objective. For simplicity, we further assume that the per-epoch reward is bounded and normalized, that is, for all h, s, a, the collected random reward $R(h,s,a) \in [0,1]$ with the mean reward denoted as $r(h,s,a)$. The initial state of $\mathcal{M}$ is denoted by $\mathbf{s}_1$. We call $\mathcal{M}$ the *target MDP*. Our goal is to find an optimal policy $\pi : \mathcal{S} \times \mathcal{H} \to \mathcal{A}$ for $\mathcal{M}$ with the objective as

$$\max_{\pi} V(\mathcal{M}, \pi) := \max_{\pi} \mathbb{E}\left[\sum_{h=1}^{H} R(h, s_h, a_h = \pi(s_h, h)) \,|\, \mathbf{s}_1\right]. \tag{1}$$

Denote the optimal policy as π^*, and the optimal Q-functions (state-action values) under π^* by $Q^*(h,s,a)$. The Bellman optimality equation characterizes Q^*:

$$Q^*(h,s,a) = r(h,s,a) + \sum_{s' \in \mathcal{S}} P(s'; h, s, a) \max_{a' \in \mathcal{A}} Q^*(h+1, s', a'), \quad \text{with } Q^*(H+1, \cdot, \cdot) \equiv 0. \tag{2}$$

This MDP model is fairly general and applies to many applications. We take preventative intervention as an example to illustrate. The state $s \in \mathbb{R}^d$ typically includes the health status of the patient and other relevant information. The intervention action $a \in \mathcal{A}(s,h) \subseteq \mathcal{A} \equiv \{0, 1, \ldots, A\}$. Both the state and action choice could be flexibly designed according to the application context. For example, a three-dimensional state $s = (s_1, s_2, s_3)$ can be employed to track a patient's health status, the number of interventions administered by far, and the number of switches between different intervention types. Following the literature, $s_1 \in \{0, 1, \ldots, M\}$ represents a risk score, where 0 typically denotes a healthy state and M corresponds to an absorbing state associated with adverse events (e.g., mortality, readmission, disengagement from mobile health programs, or completion of the preventative phase). Incorporating the number of interventions (s_2) and switches (s_3) accounts for patient fatigue or other negative effects resulting from excessive interventions or frequent switches. These factors can reduce patient adherence and diminish intervention effectiveness (Ayer et al. 2016). A common form of the cost (negative reward) is, for example, $-r(h,s,a) = c(h,s,a) = c_a(s) + c_M \cdot P(s_1 = M; h, s, a)$, where c_M is the penalty for entering M (adverse event occurring), and $c_a(s)$ is the intervention cost for action a (including no intervention) and can be state-dependent, for example, higher fatigue level makes the intervention more costly. Note that we are not necessarily assuming Markovian transition, for example, tracking (s_2, s_3) in the state leads to a history-dependent setting. Similarly, a partially observable setting could be accounted for by augmenting the state with an information state (Piri et al. 2022); we focus on the observable state setting in this paper.

3.2. Reinforcement Learning Framework Using Target and Historical Data

The MDP model introduced in Section 3.1 is for one patient type, and the optimization problem (1) can be solved via the Bellman Equation (2) if the true model parameters (R, P) are known. In real applications, however, (R, P) are not known and often highly *heterogeneous* among target patients. This requires accurately estimating $(R^{(i)}, P^{(i)})$ for each patient type i, yet the primary challenge is the limited data available for individual types. To address this small-sample issue in multistage sequential decision-making, as discussed in the introduction, we develop a new reinforcement learning (RL) framework that integrates both target (online) and historical (offline) data. We next outline the components of our RL framework and examine the feasibility of online learning in Section 7 within the context of our case study.

3.2.1. Target Patients. We do not know the parameters of the target MDP $\mathcal{M}$ a priori, namely, (R, P) are unknown (we drop the superscript i when referring to a given class of target patients for notational convenience). Instead, we collect the *target data*, denoted as $\mathcal{D}$, gradually in an online manner by selecting actions according to a policy in each iteration and interacting with $\mathcal{M}$. Throughout this process, we continuously update our estimates of (R, P) at each iteration. The goal is to learn an effective policy for the target class, with the notion of "iteration" formally defined in Section 4.

3.2.2. Augmenting with Historical Data. Our RL framework leverages offline historical data from other users/systems that are similar to $\mathcal{M}$ to facilitate the target learning. We denote the MDP for the historical patient populations as $\mathcal{M}_0 = (\mathcal{H}, \mathcal{S}, \mathcal{A}, R_0, P_0)$. The historical patients share the same state and action spaces with the target patients, and the same horizon length H. However, they differ in the reward and transition probabilities (R_0, P_0), and these parameters are also unknown. We do not interact with the historical patients during the target learning process but have access to data generated from $\mathcal{M}_0$ offline, denoted as $\mathcal{D}_0$—namely, the *historical data*. However, unlike offline algorithms that use static estimates for (R_0, P_0) derived from $\mathcal{D}_0$, we continuously update our estimates for (R, P) using both the target data $\mathcal{D}$ and the historical data $\mathcal{D}_0$.

We formalize the "closeness" between the target and historical MDPs in Assumption 1.

Assumption 1. *There exists a constant $\Delta > 0$, such that $\forall (s,a) \in \mathcal{S} \times \mathcal{A}$, $1 \le h \le H$,*

$$|r(h,s,a) - r_0(h,s,a)| \le \Delta, \quad \|P(\cdot;h,s,a) - P_0(\cdot;h,s,a)\|_1 \le \Delta.$$

We emphasize that our approach does not require any parametric assumptions about how target and historical patients are connected in terms of model parameters. Instead, we rely solely on the nonparametric Assumption 1 to measure their closeness. This distinguishes our method from regression-based approaches, which assume that target and historical patients with similar features exhibit similar outcomes (rewards) and transition probabilities—an assumption typically enforced through a parametric model. In other words, our learning framework operates in a "model-free" manner, which avoids potential model misspecification. Another benefit of our formulation is that it only requires aggregate statistics, (R_0, P_0), from the historical data set $\mathcal{D}_0$, avoiding the need to share individual-level data across healthcare organizations—an often challenging task. Additionally, the representation via (R_0, P_0) accounts for scenarios where historical data comes from a mixture of patient classes or populations, as (R_0, P_0) reflects the weighted mixture of these MDPs and can be estimated through empirical averages (Chen et al. 2020).

4. Baseline Algorithm Design and Regret Analysis

In this section, we introduce our main algorithmic framework, named the *perturbed value iteration* with weight λ (PVI-λ). The PVI-λ framework iterates over four steps in each iteration t:

1. Estimate $(\hat{R}, \hat{P})$ using a calculation procedure that weighs between the historical and target data, $\mathcal{D}$ and $\mathcal{D}_0$, with a weight function $\lambda(\cdot)$ that depends on the target data collected up to this iteration. Then, use $(\hat{R}, \hat{P})$ to estimate the Q-function, $\hat{Q}_t(h,s,a)$, as defined in (2).
2. Add some perturbation terms to $\hat{Q}_t(h,s,a)$ and obtain $\tilde{Q}_t(h,s,a)$ to encourage exploration.
3. Interact with $\mathcal{M}$ using policy $\pi_t(s,h) = \arg\max_{a \in \mathcal{A}} \tilde{Q}_t(h,s,a)$ for a complete episode of H epochs and collect new data.
4. Add the newly collected data to $\mathcal{D}$ and move to the next iteration.

In this section, we focus on the cleanest "baseline setting" to introduce the core concept of PVI-λ and the design specification. This baseline setting uses tabular representation for the Q-function for a given target patient class. We first present the meta algorithm in Section 4.1, which highlights the key steps that differ our framework from other value-based RL methods. We then carry out a theoretical analysis on the decision quality of PVI-λ, measured by the regret accumulated in given iterations, in Section 4.2. We connect the decision quality to a sequence of confidence radii, which transforms our problem of designing an efficient data-pooling RL algorithm into a more specific task: optimizing the weight function to minimize the confidence radii. In Section 4.3, we specify the optimal weight function λ^* and the resulting regret bound for the `PVI-`λ^* algorithm. This analytical framework lays the

foundation for our asymptotic analysis in Section 5 and extensions in Section 6, which include (i) handling an unknown Δ, (ii) integrating multiple historical datasets from different sources, and (iii) incorporating contextual information.

4.1. Algorithm Framework: PVI-λ

We give the complete description of the general PVI-λ framework in Algorithm 1 for the baseline tabular setting and a given target class. The weight function λ remains to be general here. PVI-λ differs from other value-based RL methods like LSVI (Bradtke and Barto 1996) in two key steps: (i) value function estimation (Step 1), and (ii) adding perturbation terms tailored to the pooled value function estimates (Step 2). We elaborate on these two steps below; Steps 3–4 are not repeated.

Step 1. We estimate the state-action value function, the Q-function, utilizing both historical and target data. In the tabular setting, this is reduced to estimate the mean rewards and transition probabilities for each (h,s,a) tuple in each iteration t. We consider a given tuple (h,s,a), and suppose there are $n_t(h,s,a)$ samples from the target data and $N(h,s,a)$ samples from the historical data for this tuple at iteration t. Note that $n_t(h,s,a)$ is random and nondecreasing in t since new target samples are gradually collected during the learning process, while $N(h,s,a)$ is a fixed constant determined by the historical data. For the simplicity of notation, when the iteration t is given, we write $n \equiv n_t(h,s,a)$ and $N \equiv N(h,s,a)$ unless otherwise specified.

Let $\overline{r}_n$ and $\overline{P}_n$ be the empirical mean reward and vector of transition probabilities obtained from the n samples of target data, and $\overline{r}_0$ and $\overline{P}_0$ obtained from the N samples of historical data. The *λ-estimates* of the mean reward r and transition probabilities P are defined as

$$\hat{r}_t^\lambda := \lambda\overline{r}_n + (1-\lambda)\overline{r}_0, \quad \hat{P}_t^\lambda := \lambda\overline{P}_n + (1-\lambda)\overline{P}_0 \tag{3}$$

for the given (h,s,a) tuple. Here, the weight λ can be a constant (e.g., $\lambda = 1$ corresponding to no-pooling), or a function in terms of n and N, $\lambda = \lambda(n,N)$, where the two sample sizes reveal the relative accuracy of target estimates, $\overline{r}_n$ and $\overline{P}_n$, in comparison with those obtained from historical data, $\overline{r}_0$ and $\overline{P}_0$. When $N = 0$, $\lambda = 1$ by default and (3) becomes the no-pooling version.

The Q-function can then be estimated using the Bellman optimality Equation (2) with the λ-estimates of the mean reward and transition probabilities, that is,

$$\begin{aligned}\hat{Q}_t^\lambda(h,s,a) &= \hat{r}_t^\lambda(h,s,a) + \sum_{s'\in\mathcal{S}} \hat{P}_t^\lambda(s';h,s,a) \max_{a'\in\mathcal{A}(s',h+1)} \hat{Q}_t^\lambda(h+1,s',a') \\ &\equiv \hat{r}_t^\lambda(h,s,a) + \hat{P}_t^\lambda(h,s,a)\cdot \max_{a'\in\mathcal{A}(s',h+1)} \hat{Q}_t^\lambda(h+1,\cdot,a')\end{aligned} \tag{4}$$

for $h = H, H-1, \ldots, 1$, with $\hat{Q}_t^\lambda(H+1,s,a) = 0$. Here, $\cdot$ represents the inner product of two vectors. Clearly, selecting an appropriate weight function is crucial for designing an efficient algorithm.

Step 2. The perturbation term addresses the exploration issue in online learning. It is known that value iteration based solely on the estimated value function could get stuck in suboptimal solutions without any exploration (Osband et al. 2016). The PVI-λ framework introduces a set of perturbation terms $\{w_t^\lambda(h,s,a)\}$ and injects them into the estimated value functions, that is,

$$\begin{aligned}\tilde{Q}_t^\lambda(h,s,a) = \hat{r}_t^\lambda(h,s,a) + w_t^\lambda(h,s,a) + \hat{P}_t^\lambda(h,s,a) \\ \cdot \max_{a'\in\mathcal{A}(s',h+1)} \tilde{Q}_t^\lambda(h+1,\cdot,a').\end{aligned} \tag{5}$$

Correspondingly, the value iteration performed via (5) uses the perturbed version $\tilde{Q}_t^\lambda(h,s,a)$ instead of $\hat{Q}_t^\lambda(h,s,a)$. The design of the perturbation $w_t^\lambda(h,s,a)$ is the second key component in the algorithm design besides choosing the weight function λ. We specify the design of both key components in Section 4.3, and we conclude this section with two remarks.

Algorithm 1 (PVI-λ)

Initialize $\tilde{r}_1 = 0$ and $\overline{P}_1 = 0$, data set $\mathcal{D} = \{\}$, $\tilde{Q}_t^\lambda(H+1,s,a) \equiv 0$, $\forall t,s,a$.
Input $\lambda = \lambda(n,N)$; $(N(h,s,a), \overline{r}_0(h,s,a), \overline{P}_0(h,s,a))$ obtained from the historical data.
for $t = 1$ **to** T **do**
  Compute $\overline{r}_t, \overline{P}_t$ from $\mathcal{D}$.
  for $h = H, H-1, \ldots, 1$ **do**
    for all (s,a) pairs **do**
      Generate a perturbation term $w_t^\lambda(h,s,a)$
      Compute the data-pooling estimators $\hat{r}_t^\lambda(h,s,a)$ and $\hat{P}_t^\lambda(h,s,a)$ according to (3) with $n = n_t(h,s,a), N = N(h,s,a)$, and $\lambda = \lambda(n,N)$.
      Compute data-pooling perturbed value function $\tilde{Q}_t^\lambda$ according to (5).
    end for
  end for
  Initialize $S_1^t = s_1$, $A_1^t = \arg\max_{a\in\mathcal{A}(s_1,1)} \tilde{Q}_t^\lambda(1,s_1,a)$.
  for $h = 2, \ldots, H$ **do**
    Sample state $S_h^t \sim P(\cdot; h-1, S_{h-1}^t, A_{h-1}^t)$.
    Select action $A_h^t = \arg\max_{a\in\mathcal{A}(S_h^t,h)} \tilde{Q}_t^\lambda(h,S_h^t,a)$.
    Collect reward R_h^t.
  end for
  Update $\mathcal{D} = \mathcal{D} \cup \{S_1^t, A_1^t, R_1^t, \ldots, S_H^t, A_H^t, R_H^t\}$
end for

Remark 1 (Meaning of Iteration). Like other value-based RL methods, each iteration t of the PVI-λ algorithm involves first updating parameter estimates and policy π_t, then applying π_t to one single patient, collecting a complete trajectory of her state evolution over H epochs. In other words, iteration t indexes entire

trajectories for different patients, not sequential observations within a patient; whereas index h refers to specific epochs within the fixed observation horizon H. We adhere to these notations throughout the paper to clearly distinguish between t and h.

Remark 2 (Recovering Benchmark Algorithms). The PVI-λ framework covers several commonly used RL algorithms. For example, when $\lambda \equiv 1$, the PVI-1 algorithm corresponds to the classic `no-pooling` algorithm using only the target data. In addition, with proper selection of $w_t^1(h,s,a)$, PVI-1 covers two popular classes of RL methods: `UCB-based` and `TS-based` RL algorithm, where UCB stands for Upper-Confidence-Bound, and TS stands for Thompson-Sampling. The TS-based version is also known as the Randomized Least-Squares Value Iteration (RLSVI), initially developed in Wen (2014) and extended in Osband et al. (2016), Russo (2019). Another example is when we set the weight as $\overline{\lambda} \equiv n/(n+N)$. Then PVI-$\overline{\lambda}$ corresponds the `naive-pooling` algorithm, that is, we simply treat the historical patients as if they follow the same distribution as the target patients and merge the data for parameter estimation.

4.2. Decision Quality of PVI-λ

In this section, we formally define our primary metric, regret, and conduct a theoretical analysis of the regret for the general PVI-λ algorithm. This analysis serves as a vital foundation for linking the two key components of the algorithm design, namely the weight function and perturbation term, to the decision quality. It also serves as the basis for the subsequent development in Section 4.3.

The (total) regret accumulated in T iterations, defined in (6), measures the difference between the maximum value achieved under the true optimal policy π^* and the expected reward collected when decisions are made by the RL algorithm without knowing the true parameters:

$$Regret(T) := \sum_{t=1}^{T} (V(\mathcal{M},\pi^*) - \mathbb{E}[V(\mathcal{M},\pi_t)]), \qquad (6)$$

where π_t is the policy prescribed by the RL algorithm in iteration t. In implementing the PVI-λ algorithm, π_t selects actions based on the perturbed value function $\tilde{Q}_t^{\lambda}$, which leverages both historical and online target data. As a result, in addition to the usual randomness arising from the interaction between the algorithm and the generation of online data (including the perturbation terms), the expectation here also accounts for the uncertainty associated with the generation of historical data of sample size N.

An RL algorithm is more efficient if it has a smaller regret. Reducing regret is important not only from the technical perspective but also for practical implications. In preventative intervention, efficiently allocating limited resources to patients who need them most is critical for system-wide improvements. Nonmedical interventions, though nonintrusive, can lead to inefficiencies and wasted resources if learning is inefficient, such as unnecessary text messages. Thus, minimizing regret is essential. Using regret as our decision quality metric also highlights a key difference from MDP and offline algorithms. While MDPs and offline methods focus on (in-sample) cost minimization, our approach considers the costs incurred during data collection in new patients, making it more applicable to dynamic and evolving environments.

Next, we establish the regret bound for the general PVI-λ algorithm and make two observations: (i) the regret bound largely depends on the *confidence radii* of the parameter estimators $\hat{r}_t^{\lambda}$ and $\hat{P}_t^{\lambda}$ for $1 \le t \le T$; (ii) the standard deviation of the perturbation term, w_t^{λ}, should be proportionate to the confidence radius in each iteration t. These connections are intuitive. For (i), the decision quality is directly influenced by the accuracy of the value function estimation, which, in turn, relies on estimates of the MDP parameters. In other words, the algorithm can make better decisions when the parameters and value function are estimated with greater accuracy. For (ii), the perturbation must compensate the estimation error in the value function in order to ensure sufficient exploration.

Definition 1 (Confidence Radii). Given $\delta > 0$, we say that a sequence of functions $\{\varepsilon^{\lambda}(n,N) : 0 \le n \le T, N \ge 0\}$ are the confidence radii for estimators $\{\hat{r}_t^{\lambda}, \hat{P}_t^{\lambda} : 1 \le t \le T\}$ with confidence level $1-\delta$, if for each (h,s,a) tuple with $N(h,s,a) = N$ and iteration $t(n) \in \{0 \le t \le T : n_t(h,s,a) = n\}$,

$$\begin{aligned}
&\mathbb{P}(|\hat{r}_{t(n)}^{\lambda}(h,s,a) - r(h,s,a)| > \varepsilon^{\lambda}(n,N)) < \delta/(HSAT),\\
&\mathbb{P}(\|\hat{P}_{t(n)}^{\lambda}(h,s,a) - P(h,s,a)\|_1 > 2\sqrt{S}\cdot\varepsilon^{\lambda}(n,N)) < \delta/(HSAT),\\
&\mathbb{P}(|\hat{r}_{t(n)}^{\lambda}(h,s,a) - r(h,s,a) + \langle \hat{P}_{t(n)}^{\lambda}(h,s,a) - P(h,s,a), V_{h+1}^*\rangle|\\
&> H\cdot\varepsilon^{\lambda}(n,N)) < \delta/(HSAT),
\end{aligned} \qquad (7)$$

where $V_{h+1}^*(\cdot) = \max_{a\in\mathcal{A}} Q^*(h+1,\cdot,a) \in \mathbb{R}^S$ is the optimal value-to-go from horizon $h+1$.

When $\lambda = 1$, $\{\hat{r}_t^1, \hat{P}_t^1 : 1 \le t \le T\}$ are estimated using only target data, according to the stack-of-reward model for the standard no-pooling tabular RL (see lemma 2 of Russo 2019). Hoeffding's inequality yields the following expression for the confidence radii

$$\left\{\varepsilon^1(n,N) \equiv \sqrt{\log(2HSAT/\delta)/(2n)} : 1 \le n \le T\right\}$$

for the no-pooling estimators $\{\hat{r}_t^1, \hat{P}_t^1 : 1 \le t \le T\}$ with confidence level $1-\delta$ and $n \ge 1$. We define $\varepsilon^{\lambda}(0,N) \equiv 1$

as the trivial upper bound for all λ. In the following regret analysis, we fix $\delta = 1/(HT)$.

For PVI-λ, sufficient exploration needs to be injected to effectively learn the optimal policy. In particular, the standard deviation of the perturbation term $w_t^\lambda(h,s,a)$ should be set in proportional to the confidence radius:

$$w_t^\lambda(h,s,a) = H\varepsilon^\lambda(n_t(h,s,a), N(h,s,a)) \cdot \xi_t(h,s,a), \tag{8}$$

where $\xi_t(h,s,a)$ are i.i.d. copies of certain random variables, whose distribution is chosen according to whether we use `UCB`- or `TS`-based algorithm; see Appendix EC.1. Under the specification in (8) and some mild technical conditions, the regret of the PVI-λ can be bounded by a function of the confidence radii $\{\varepsilon^\lambda(n,N)\}$, outlined in Theorem 1.

Theorem 1 (Informal). *Suppose the perturbation terms $w_t(h,s,a)$ are generated according to (8). Then, under certain technical conditions, the regret of PVI-λ with T iterations, $\text{Regret}^\lambda(T)$, is bounded by*

$$RB^\lambda(T) \equiv K_1 \max\left\{ \sum_{h,s,a} \sum_{n=0}^{n_{h,s,a}-1} \varepsilon^\lambda(n, N(h,s,a)) : n_{h,s,a} \geq 0, \sum_{s,a} n_{h,s,a} = T,\ \forall h \right\} + K_2, \tag{9}$$

where $\varepsilon^\lambda(\cdot,\cdot)$ are the confidence radii for $\{\hat{r}_t^\lambda, \hat{P}_t^\lambda : 1 \leq t \leq T\}$ with confidence level $1 - 1/(HT)$, and constants $K_1, K_2 > 0$ are independent of T.

The formal statement of Theorem 1, including the expressions of constants K_1, K_2, and its proof are specified in Appendix EC.2.1. Note that n in the summation is a dummy variable, which enumerates over all possible target sample sizes through the learning procedure. This is slightly different from $n = n_t(h,s,a)$ used in earlier analysis, where the tuple (h,s,a) and iteration t are fixed.

The bound in (9) applies to any given number of historical samples. If for certain (h,s,a) tuple, $N(h,s,a) = 0$, then the estimates become the no-pooling version and so as the confidence radii. In the special case where all historical tuples are *balanced sampled* with $N(h,s,a) \equiv N$ (e.g., through randomized controlled trials), we can simplify the general form in (9) into

$$RB^\lambda(T) = HSAK_1 \sum_{n=0}^{\lfloor T/(SA) \rfloor} \varepsilon^\lambda(n,N) + K_2 \tag{10}$$

using the pigeonhole principle. We leave the derivation of (10) to Appendix EC.2.1.

4.3. Adaptive-Pooling with Decision-Based Weight Design

Given the specification in Equation (8) for the perturbation term, the primary task is to determine an appropriate weight function design. In this section, we introduce our `adaptive-pooling` algorithm, PVI-λ^*, which selects a sequence of weight functions $\lambda^*(n,N)$ to minimize the confidence radii and is adaptive during the learning procedure. The idea for this design stems from the theoretical analysis in Section 4.2, where Theorem 1 establishes the dependence of the regret bound on the confidence radii $\{\varepsilon^\lambda(n,N)\}$. Thus, by minimizing the confidence radii through the determination of $\lambda^*(n,N)$, we can effectively reduce the regret and improve the decision quality. To formalize this intuition, we first specify the optimal λ^* in Proposition 1 and then present results on the regret bound improvement. The proofs are in Appendix EC.2.2 –EC.2.4.

Proposition 1. *Let $\delta > 0$ be a given confidence parameter. Suppose for the given (h,s,a) tuple, we have n target sample and N historical sample. Under Assumption 1,*

1. *Given a weight function $\lambda(n,N)$, the sequence of functions*

$$\varepsilon^\lambda(n,N) \equiv \begin{cases} \sqrt{\log(2HSAT/\delta)\left(\dfrac{\lambda(n,N)^2}{2n} + \dfrac{(1-\lambda(n,N))^2}{2N}\right)} + \Delta(1-\lambda(n,N)) & n \geq 1, \\ 1 & n = 0, \end{cases} \tag{11}$$

are confidence radii for $\{\hat{r}_t^\lambda, \hat{P}_t^\lambda : 1 \leq t \leq T\}$.

2. *Given a pair of (n,N), the function $\varepsilon^\lambda(n,N)$ obtains its minimum value at*

$$\lambda^*(n,N) \equiv \begin{cases} \dfrac{n + Nn\Delta/\sqrt{(N+n)\log(2HSAT/\delta)/2 - \Delta^2 Nn}}{N+n} & \text{if } n < \log(2HSAT/\delta)/(2\Delta^2), \\ 1 & \text{otherwise.} \end{cases} \tag{12}$$

The expression of λ^* in (12) depends not only on the sample sizes n and N but also Δ, which is assumed to be a known constant in the baseline setting. We discuss how to deal with unknown difference gap Δ in Section 6.1. This weight λ^* is *adaptive* to the learning procedure through the changing $n = n_t(h,s,a)$ as iteration t changes. In particular, λ^* increases with n and Δ, which implies that we place greater trust in the estimates from the target data and rely on historical data

less as (i) more target data are gathered, and (ii) the difference between historical and target patients is large. When this difference is substantial or sufficient target data has been collected, λ^* reaches 1, reducing the data-pooling estimators to the no-pooling estimators that use only target data. Conversely, when $\Delta = 0$, meaning historical and target data are from the same population, we obtain $\lambda^* = \overline{\lambda}$, recovering the naive-pooling estimators. Beyond its interpretability, the closed-form expression of λ^* also significantly improves computational efficiency.

Let $RB^1(T)$ and $RB^{\overline{\lambda}}(T)$ be the regret bounds of two benchmarks, `no-pooling` and `naive-pooling`, respectively, and $RB^{\lambda^*}(T)$ be the regret bound of our `adaptive-pooling` algorithm, which sets the weight as λ^* given in Proposition 1. As discussed in Remark 2, the two benchmarks set the weight as 1 and $\overline{\lambda} = n/(n+N)$, respectively. The perturbation term in each algorithm follows (8) using the corresponding weight function. Since the regret bounds increase with the confidence radii, as a direct consequence of Theorem 1 and Proposition 1, the `adaptive-pooling` algorithm, with λ^* designed to minimize the confidence radii, achieves a smaller regret bound.

Corollary 1. *Under the same conditions of Theorem 1,*

$$RB^{\lambda^*}(T) \le RB^1(T), \quad RB^{\lambda^*}(T) \le RB^{\overline{\lambda}}(T).$$

Remark 3. Corollary 1 holds even when $N(h,s,a) = 0$ for certain (h,s,a) tuples. As long as historical data are available for some tuples visited during learning, utilizing the historical data helps reduce the corresponding confidence radii, thereby lowering the regret bound, which takes the maximum over the sum of all confidence radii throughout the learning process. In the extreme case where $N(h,s,a) = 0$ for all (h,s,a) tuples, $\lambda^* = \overline{\lambda} = 1$ by definition, and the regret bounds are all equal.

5. Asymptotic Analysis

In this section, we establish the asymptotic setting to characterize the reduction in the regret bound relative to two benchmarks and to more precisely quantify the improvement achieved by the `adaptive-pooling` algorithm beyond the results in Corollary 1. Specifically, we analyze a *small-data regime* limit regime, where the historical data significantly outweighs the target data, and the difference gap between the two populations is relatively small. This analysis enables us to derive explicit asymptotic ratios between RB^{λ^*}, $RB^{\overline{\lambda}}$, and RB^1.

5.1. Asymptotic Setup

We consider a sequence of problem instances indexed by $N \ge 1$, where all instances share the same target MDP model $\mathcal{M} = (\mathcal{H}, \mathcal{S}, \mathcal{A}, R, P)$. Let $\mathcal{M}_0^N$ denote the MDP model that generates the historical data set in the N-th instance. As $N \to \infty$, we increase the amount of historical data and the total number of iterations in target learning, while gradually reducing the difference gap between the historical data and the target MDP. Specifically, we impose the following conditions on the historical data size, the number of iterations, and the difference gap.

Assumption 2 (Small-data Regime). *In the N-th environment,*

1. *For each tuple of (s,a,h), there are N observations in the historical data.*
2. *The total number of iterations $T(N) = c_1 SA\sqrt{N}$ for some constant $c_1 > 0$.*
3. *The square of the difference gap $\Delta^2(N) = c_2 N^{-1/2}\log(N)$ for some constant $c_2 > 0$.*

The assumption that all tuples are balanced sampled in the historical data set is mainly for a closed-form expression of the regret bound (see also the discussion after Theorem 1). The assumption implies that the total number of historical observations is $HSAN$, whereas the total number of target observations is on the order of $O(HSA\sqrt{N})$ by the end of T iterations. As N approaches infinity, the quantity of historical data significantly surpasses that of the target data, which is why we refer to this regime as the "small-data regime." Meanwhile, as the number of iterations increases, more data are collected from the target patients. This leads to a decrease in the standard deviation of the estimated mean rewards and transition probabilities from the target data (decreasing at a rate proportional to the square root of the sample size). Thus, we let the bias from the historical data, Δ, shrink at the same rate as the standard deviation of the estimated parameters from the target data, with some additional logarithmic terms, that is, $\Delta(N) = \tilde{O}(T(N)^{-1/2})$. The rationale is that, if Δ were to remain constant, the bias would eventually dominate the benefits of variance reduction from data pooling. By scaling Δ with N, we ensure the bias diminishes relative to the variance reduction, making the theoretical results meaningful. This scaling is similar to the moderate signal regime discussed in Kuang and Wager (2024).

5.2. Asymptotic Results

For the asymptotic analysis, we denote the regret bounds of the three algorithms, `no-pooling`, `naive-pooling`, and `adaptive-pooling`, when applying them to the N-th instance for $T(N)$ iterations, as $RB^1(N)$, $RB^{\overline{\lambda}}(N)$, and $RB^{\lambda^*}(N)$, respectively. We establish the following asymptotic results for the ratios of these regret bounds.

Theorem 2. *Let $c_0 = c_1 c_2$. Then, for all $N \ge 0$ and $c_0 > 0$, $RB^{\lambda^*}(N) \le \min(RB^1(N), RB^{\overline{\lambda}}(N))$. Additionally, under*

Assumptions 1 and 2, the ratios

$$\limsup_{N\to\infty}\frac{RB^{\lambda^*}(N)}{RB^1(N)} \le \begin{cases} \frac{\sqrt{c_0}}{2}, & c_0 \le 1 \\ 1-\frac{1}{2\sqrt{c_0}}, & c_0 > 1, \end{cases}$$
$$\limsup_{N\to\infty}\frac{RB^{\lambda^*}(N)}{RB^{\bar{\lambda}}}(N) \le \begin{cases} 1, & c_0 \le 1 \\ \frac{2\sqrt{c_0}-1}{c_0}, & c_0 > 1. \end{cases} \tag{13}$$

The proof of Theorem 2 is in Appendix EC.2.4.

To interpret Theorem 2, note that the constant c_0 measures the relative importance of target data in the learning process under different parameter scenarios. A higher value of c_0 indicates a larger sample size of target data and/or more significant difference between the historical and target data. Consequently, the target data should play a more dominant role in learning. As $c_0 \to \infty$, meaning the target data becomes increasingly important, the regret bound ratio of the adaptive-pooling algorithm over the naive-pooling algorithm tends to zero, specifically $(2\sqrt{c_0}-1)/c_0 \to 0$. This implies that the adaptive-pooling algorithm can achieve significantly better performance, in terms of the regret bound, compared with the naive-pooling algorithm, which overly relies on historical data and fails to account for differences between historical and target patients. This highlights a key mechanism of our weight design: λ^* effectively balances the *bias-variance tradeoff*. In particular, the term $\sqrt{\left(\frac{\lambda(n,N)^2}{2n}+\frac{(1-\lambda(n,N))^2}{2N}\right)}$ in Equation (11) accounts for variance reduction by introducing the N historical data, while the term $\Delta(1-\lambda(n,N))$ accounts for bias.

Conversely, not using historical data at all also pays a price in certain regime. As c_0 approaches zero, indicating that the historical data are highly informative, the regret bound ratio of the adaptive-pooling algorithm over the no-pooling algorithm tends to zero, specifically $\sqrt{c_0}/2 \to 0$. This suggests that the adaptive-pooling algorithm can achieve significantly better performance, in terms of the upper bound of regret, than the no-pooling algorithm.

Our results also illustrate when adaptive-pooling reduces to naive-pooling or no-pooling. When $c_0 < 1$, that is, the gap between historical and target data are small or the total number iteration is small such that the impact of variance dominates, the performance of adaptive-algorithm is similar to naive-pooling. On the other hand, when $c_0 \to \infty$, that is, the gap is large or the total number of iteration is large so that the impact of bias dominates, the performance of adaptive-algorithm becomes similar to no-pooling. This observation is aligned with the discussion on λ^* in Section 4.3.

We conclude this section with the following remark. The theoretical results in Theorem 1, Corollary 1, and Theorem 2 primarily provide upper bound comparisons among the algorithms. We note that the regret bound might not be tight—in the RL literature, algorithms with provable tight bounds beyond the bandit setting usually require additional assumptions. *Exact regret* characterization is an important and emerging area of research recently, but transitioning from upper bound analysis to exact regret analysis involves significant complexity. Leveraging results from Kalvit and Zeevi (2021) and using the same asymptotic framework (with slightly different assumptions), we have derived asymptotic limits for the exact regret in a specific two-armed bandit setting. Let $R^{\lambda^*}(N)$ denote the expected regret of the adaptive-pooling algorithm and $R^1(N)$ that of the no-pooling algorithm. We show that

$$R^{\lambda^*}(N)/R^1(N) \to 0 \text{ as } N \to \infty.$$

This result shows that the exact regret of the adaptive-pooling algorithm can become arbitrarily smaller than that of no-pooling, which provides consistent theoretical insights as those in Theorem 2. The difficulty in the proof of this result lies in that exact regret analysis requires comparing *differences* in confidence radii between the two arms (unlike characterizing how confidence radii decrease with data-pooling estimates in the upper bound analysis). This requires precise characterization of both confidence radii and noise terms, accounting for noise in both target and historical data sources. Techniques such as the Law of the Iterated Logarithm are needed to analyze estimation errors in the data. We refer readers to the technical companion (Anonymous 2025) for a formal statement and proof. Due to the significant challenges, recent advances, such as Kalvit and Zeevi (2021) and Kuang and Wager (2023), have primarily focused on the bandit setting and by far there is no uniform framework for handling noise terms. Extension of the exact regret analysis in the two-armed bandit setting to the full RL requires substantial new development and is left for future work. Meanwhile, the primary purpose of our regret bound analyses is to guide the design of data-pooling weights that effectively improve performance. The empirical success of our algorithm in significantly reducing regret compared with benchmarks, as shown in Section 7, aligns with these theoretical insights even they are based on upper bound results.

6. Practical Implementations of PVI-λ

The theoretical analysis in Section 4 is performed in the baseline setting, which assumes tabular learning for one target class. It also assumes a single historical data set and known difference gap Δ. These assumptions may impose barriers to implementing our algorithm in practice. In this section, we discuss how to

implement our adaptive-pooling algorithm in more practical settings. We address how to handle (i) the unknown difference gap Δ; (ii) multiple historical datasets; and (iii) potential heuristics for incorporating contextual information during target learning, while highlighting the key differences between contextual bandit learning and RL.

6.1. Unknown Difference Gap

The optimal weight $\lambda^*(n,N)$ given in Proposition 1 requires we know the difference gap Δ a priori. When Δ is not known exactly in practical implementation, we use an estimated Δ. In each iteration t, we estimate Δ using the difference between the current estimated $(\overline{r}_t, \overline{P}_t)$ and $(\overline{r}_0, \overline{P}_0)$. Specifically, for given (h,s,a) tuple and iteration t, we compute

$$\Delta_t^{hsa} \equiv \max(|\overline{r}_0(h,s,a) - \overline{r}_t(h,s,a)|, \|\overline{P}_0(\cdot;h,s,a) - \overline{P}_t(\cdot;h,s,a)\|_1).$$

We then plug $\Delta = \Delta_t^{hsa}$ into (12) to compute the weight λ^* and the estimators $\hat{r}_t^{\lambda^*}(h,s,a)$ and $\hat{P}_t^{\lambda^*}(h,s,a)$. We have performed numerical experiments, in which we compare this estimated approach with the one that uses the groundtruth Δ. We find that they produce similar performance and beat other benchmarks. Moreover, to test the robustness of this approach, we also perturb the value of the estimated Δ_t^{hsa} in EC.4.7.1. The results are close within a relatively wide range, implying the robustness to the choice of Δ and the practical implementability of our approach.

6.2. Multiple Historical Groups

In practice, it is common that multiple sets of historical data are available from different sources (populations), for example, hospitals at different locations. Their similarities to the target MDP could be different. It is possible to choose different weight functions for each of the historical datasets to further improve the decision quality over our baseline algorithm. Assume that there are K sets of historical data indexed by $k = 1,2,\ldots,K$. Denote by $\mathcal{M}_{0k} = (H, \mathcal{S}, \mathcal{A}, R_{0k}, P_{0k})$ the MDP model that generates historical data set k. We assume that the decision-maker has the following prior knowledge on the similarity between the target MDP $\mathcal{M}$ and each of the MDP $\mathcal{M}_{0k}$.

Assumption 3. *For each $k = 1,2,\ldots,K$, there exists a constant $\Delta_k > 0$ such that $\forall\ (s,a) \in \mathcal{S} \times \mathcal{A}$, $1 \le h \le H$,*

$$|r(h,s,a) - r_{0k}(h,s,a)| \le \Delta_k, \quad \|P(\cdot;h,s,a) - P_{0k}(\cdot;h,s,a)\|_1 \le \Delta_k.$$

Given a (h,s,a) tuple, let $\overline{r}_{0k}$ and $\overline{P}_{0k}$ be the sample mean reward and empirical transition probabilities estimated from the N_k observations in the historical data set k. Recall that for each iteration t, $n = n_t(h,s,a)$ is the number of observations from the target data, and $\overline{r}_n$ and $\overline{P}_n$ are the sample mean reward and empirical transition probabilities. Let $\mathbf{\Delta} = (\Delta_1,\ldots,\Delta_K)$ and $\mathbf{N} = (N_1,\ldots,N_k)$. For each historical group k, we use θ_k to denote its weight in the data-pooling estimates and define $\boldsymbol{\theta} = (\theta_1,\ldots,\theta_k)$. The corresponding data-pooling estimators are

$$\hat{r}_t^{\boldsymbol{\theta}} := (1 - \theta_1 - \ldots - \theta_K)\overline{r}_n + \sum_{k=1}^{K} \theta_k \overline{r}_{0k},$$
$$\hat{P}_t^{\boldsymbol{\theta}} := (1 - \theta_1 - \ldots - \theta_K)\overline{P}_n + \sum_{k=1}^{K} \theta_k \overline{P}_{0k}. \tag{14}$$

Following a similar proof for Proposition 1, we obtain the confidence radius as a function of $\boldsymbol{\theta}$:

$$\varepsilon^{\boldsymbol{\theta}}(n,N_1,\ldots,N_k) = \sqrt{\log(2HSAT/\delta)\left(\frac{(1-\theta_1-\ldots-\theta_K)^2}{2n} + \sum_{k=1}^{K}\frac{\theta_k^2}{2N_k}\right)} + \sum_{k=1}^{K}\Delta_k\theta_k.$$

As a consequence, a natural choice of the optimal weight $\boldsymbol{\theta}^*(n,N_1,\ldots,N_K)$ is

$$\boldsymbol{\theta}^*(n,N_1,\ldots,N_K) \in \arg\min_{\boldsymbol{\theta}\in\mathcal{B}} \varepsilon^{\boldsymbol{\theta}}(n,N_1,\ldots,N_K),$$
$$\mathcal{B} = \left\{\boldsymbol{\theta} : \theta_k \ge 0, k = 1,2,\ldots,K, \sum_{k=1}^{K}\theta_k \le 1\right\}. \tag{15}$$

The modification to Algorithm 1 to accommodate K groups is straightforward by using the group-specific weights $\boldsymbol{\theta}^*$. Although we do not have a closed-form expression for $\boldsymbol{\theta}^*$, we show in Appendix EC.2.5 that $\varepsilon^{\boldsymbol{\theta}}(n,N_1,\ldots,N_k)$ is a convex function on the convex set $\mathcal{B}$ and thus one can find $\boldsymbol{\theta}^*$ via any standard convex optimization solver. We also prove there that, when the difference gap Δ_k's are known, applying $\boldsymbol{\theta}^*$ yields *smaller* confidence radii than using the base-case (scalar) weight λ^*, where λ^* is calculated using the mixture of the K historical datasets. Lastly, when Δ_k's are not known in practical implementation, we can use a similar procedure as outlined in Section 6.1 to estimate Δ_k's.

6.3. Incorporating Contextual Information

The tabular learning used in the baseline setting does not use data collected from other target patients, who might share similarities across the entire target population (e.g., patients within the same new hospital). One possible extension to our baseline PVI-λ framework is to incorporates contextual information into target learning. For example, using a linear approximation of the Q-function $Q(\cdot)$ for target patients would enable

cross-learning across different target classes. In this case, a λ-estimate of the Q-function could be computed directly, treating it similarly to the λ-estimate of the reward; see Appendix EC.4.4 for implementation details of this heuristic. However, rigorously deriving the optimal weight λ_Q^* for the Q-function requires a more in-depth analysis of function approximation errors. The key challenge is that Q-function estimation is inherently recursive—it requires considering transitions to future states and involves the *max* operator (see (5)). This makes it fundamentally different from the simpler reward estimation in contextual bandits, where linear function approximations can be applied directly to the immediate reward without recursive updates and the nonlinear *max* operator. Analytical extensions of our framework to contextual RL are beyond the scope and we leave them as important future directions.

7. Case Study: Post-Discharge Intervention Planning

We demonstrate the empirical success of our data-pooling RL algorithm via a specific healthcare problem: postdischarge intervention to reduce hospital readmissions. We specify the context in Section 7.1. We describe the evaluation platform calibrated from real data in Section 7.2. We present the benchmark algorithms in Section 7.3 and compare our data-pooling algorithms with the benchmarks in Section 7.4. We discuss insights for practical implementation in Section 7.5.

7.1. MDP Setting

Patients who fail to comply with the medical advice given upon discharge (e.g., forget to take medications as prescribed) are not only contributing to millions of dollars in avoidable healthcare costs and increased hospital congestion, but are also putting themselves at higher risk for adverse events that land them back in the inpatient ward (Bresnick 2016). Follow-ups via text messages, calls or home visits to increase patients' medical adherence have proven effective in reducing hospital readmissions (Yiadom et al. 2020, Takchi et al. 2020). After a patient is discharged, the hospital makes a plan for whether to follow up with the patient at each "check point", typically each week (Liu et al. 2018). This plan is carried out by a case manager, who may dynamically adjust the plan given the state of the patient. However, due to the additional staffing requirements for the follow-ups, such interventions could result in an overall net loss if not properly designed (Chan et al. 2024).

We model this postdischarge planning problem as an MDP. Each checkpoint is a decision epoch, and without loss of generality, each week is an epoch. We consider a 30-day readmission window, hence 4 decision epochs in total; extension to the 60-day or 90-day readmissions is straightforward. The action set $\mathcal{A} = \{0,1,2\}$, with 0 as no intervention, 1 as phone calls, and 2 as home visits (stronger level than phone calls). We consider a constrained setting where we impose a maximum number of allowed intervention switches L. We track a three-dimensional state $s = (s_1, s_2, s_3)$, where $s_1 \in \{0,1\}$ corresponds to the healthy (0) and readmitted (1) state; $s_2 \in \{0,1,\ldots,L\}$ tracks the number of allowed *remaining* switches in intervention types on this patient by far, with $s_2 = 0$ being no more switch is allowed; s_3 records the previous intervention action type. As discussed in Section 3.1, s_2 is included to avoid the negative consequences due to too frequent switches of the interventions, while to maintain the Markovian property, s_3 is required to be tracked. Consequently, the feasible action depends on the current state s. That is, if $s_2 > 0$ for the current epoch $h > 0$, the feasible action set is $\mathcal{A}$; otherwise the action is only allowed to be the same as s_3 (the previous action), since $s_2 = 0$ means that the no further intervention change is allowed. The negative reward $-r(h,s,a) = c(h,s,a) = c(a) + c_R \cdot P(s_1' = 1 | h,s,a)$ for $s_1 \neq 1$, where $c(a)$ is the intervention cost with $c(0) = 0$, and c_R is the penalty cost for readmission. The detailed transition probabilities are specified in Appendix EC.3.

We note that the interventions considered here are noninvasive, as are typical ones in preventative healthcare. Examples such as phone calls or visits generally relate to administrative or informational processes and do not directly influence a patient's physiological well-being. While the exploration phase and initial inaccuracies in parameter estimation are inherent in the learning process, they do not present direct physical risks to patients, unlike experimenting with medication dosages or medical procedures. Even if the system may not initially pinpoint the most optimal strategy, it poses no immediate health hazard; instead, it refines decision-making based on observed feedback (e.g., fatigue leads to less adherence and reduced intervention benefits). Therefore, online learning is a suitable tool in our case study setting. Furthermore, by imposing constraints on the maximum allowable switches, we can mitigate the undesirability of exploring "extreme" policies involving excessive interventions or frequent switches.

7.2. Simulation Platform

We build a simulation platform to act as the "oracle" that generates data observations and evaluates the performance of different policies. To calibrate the simulation platform, we use a real data set from a large teaching hospital in Asia and calibrate the baseline setting. Building upon that, we further create several synthetic settings.

The data set contains 21,596 records of patients discharged from July 2010 to February 2011. We focus on the 30-day readmission rate as the main outcome variable. There are 21 feature variables, including basic demographics (e.g., age, gender), medical information (e.g., medical specialty, number of prior visits), insurance and diagnostic information, and intervention records (whether follow-up was scheduled upon discharge). However, due to the limitation in the data collection process, we do not know the exact number of interventions and/or types performed on each patient other than the binary label. As a result, we cannot estimate the transition probabilities depending on s_2 and s_3. Thus, when calibrating the simulation model with the real data, we take out s_2 (and s_3 accordingly) and only keep s_1 in the state. We consider $a \in \{0,1\}$ for the intervention choice, and simplify the MDP setting accordingly. We compensate this gap by creating the synthetic setting in Section 7.4.4.

7.2.1. Baseline Calibration with Real Data. We use the 9,766 samples from July to October 2010 as the historical data, and calibrate a simulator for patients discharged between November 2010 and February 2011, that is, the target patients. The key to building the simulator lies in calibrating the transition probabilities, that is, the readmission probability $P(s_1' = 1 | h, s_1 = 0, a, x)$ for different feature combinations x in each week h with or without the intervention. The main challenge in calibrating these probabilities lies in estimating the treatment effect under *selection bias*. In the historical data, patients were not randomly selected for follow-up intervention; hospitals prioritized those perceived to be at higher readmission risk. Consequently, despite receiving the intervention, these patients often had higher baseline readmission rates than those not selected. To correct for the selection bias, we apply the Heckman's two-step correction (Heckman 1976); see Appendix EC.4.

The performance of our two-step regression is measured by the out-of-sample Area Under the Curve (AUC), which is 0.71. This AUC performance is comparable with those reported in the literature for readmission prediction. The estimated treatment effect is significant, along with the following predictive features: age, number of transfers, whether admitted from the emergency department (ED), number of prior visits, the Charlson comorbidity score, had operations or not, and the length-of-stay. This is consistent with findings in the medical literature on predictors of readmission risk, such as the LACE score (Robinson and Hudali 2017). We use all of the data for a large sample size to train the prediction model, but we include the "month" feature to differentiate target versus historical patients when constructing the simulator. Therefore, the transition probabilities P_0 and P are still significantly different between the historical and target cohorts. For target patients, we assume that their feature x, which influences the readmission probability in the simulator, is observed along with the state. When implementing the tabular version of our algorithm or tabular benchmarks, we classify the target patients into 146 classes based on the predictive features listed in Table 1 to ensure similar transition probabilities and optimal policies within each class. For historical patients, we pass their feature information for benchmark algorithms that require accessing such information. Our data-pooling algorithms do not require the historical feature information; instead, we categorize historical patients into eight groups and pass the empirical averages of each group. These partitions are detailed in EC.4.2 and EC.4.3.

We assume that the intervention cost and readmission cost, $c(1) = 0.13$ and $c_R = 10$, respectively.[2] Table 1 provides a comparison of the optimal policy from the historical patients and target patients under this cost setting, showing (i) the heterogeneity among target patients with different features and (ii) the heterogeneity between target and historical patients. These results highlight the significant differences between patients, confirming the necessity of tailored intervention planning. Note that in the baseline, we differentiate historical and target cohorts by time to reflect potential shifts in patient demographics over time. In the sensitivity analysis, we explore a scenario where we distinguish the target and historical cohorts by medical specialty; see Appendix EC.4.7.3.

7.3. Benchmark Algorithms and Evaluation Framework

We use the simulator calibrated with real data to compare the performance of our data-pooling algorithm and other benchmark algorithms. To evaluate a given algorithm, we run 100 replications with 50 iterations, where each iteration is equivalent to one week. In each week, new patients are generated according to the empirical weekly arrival rates with features sampled from the empirical distribution. The simulator then generates observations for the patients: new state evolved according to the previous state and the action prescribed by the given algorithm in that week. Different from the theoretical analysis, we consider here a more practical setting of policy updating and target data collection. That is, instead of waiting for the entire episode of $H = 4$ weeks to be completed, we do a weekly update with one week (1/4 of the episode) corresponding to an iteration. Moreover, we prescribe policies to all the patients that are in the system, including (i) those who just arrived in this week, and (ii) those who have not finished all four weeks yet. At the end of the week, we collect observations from all these patients and add them to the target data set. We

Table 1. Representative Features for Target and Historical Patients, the Sample Size, and the Optimal Policy for each Class of Feature Combinations (1 = Intervene; 0 = No)

	Feature combination						Hist Pat.		Target Pat.	
Admission source	Age	#Transfers	#Prior visits	Had operations	Charlson score	Length of stay	#Sample	Opt policy	#Sample	Opt policy
NonED	[41,60]	≥1	0	1	0	≥2	197	1100	295	0000
ED	[61,80]	≤1	0	0	>0	1	241	1111	220	1100
NonED	[18,40]	≥1	0	1	>0	0	235	1100	198	0000
ED	[18,60]	≥0	0	0	≥0	≥1	323	1111	243	1100

Notes. The optimal policy is calculated using the average transition probabilities from patients in each class. The length of stay is measured in days; source of admission is emergency department admitted or not; other features are self-explanatory.

then use the given algorithm to update the MDP parameters and policies, which will be used to prescribe actions for all patients in the next week. The historical data remains unchanged. We report the average performance across the 100 replications and all target patients. Note that when calculating the regret as given in (6), we can define π^* as either (i) the optimal policy solved for each patient with feature x, or (ii) the average optimal policy for each partitioned target class, using the class's average transition probabilities and rewards. The difference between these two approaches is small (less than 3% across all tested algorithms), and we adopt the latter for simplicity.

We consider both the baseline tabular version (`adaptive-pooling tabular`) and the contextual extension discussed in Section 6.3 (`adaptive-pooling contextual`). For the latter, the patient feature x is used as the individual contextual information for target learning, allowing the algorithm to be personalized to different patients. For the tabular version, we use the predictive features to classify the target patients into 146 "target classes" and do the learning for each class separately. See Appendix EC.4.3 for the classification. Both algorithms only require the empirical averages from the historical patients given the model-free design. We use the TS-based RL as our base algorithm when designing the perturbation terms.

7.3.1. Benchmark Algorithms. We compare the performance of our two data-pooling algorithms with a few benchmarks in the following two categories.

1. Tabular-based methods:
   - The JS method applies the James-Stein (JS) estimator (Efron and Hastie 2016) to estimate the reward and transition probability by merging the historical and target data. We adapt this method for the online setting with parameter update and exploration incorporated.
   - The no-pooling tabular method uses the standard tabular RLSVI without pooling.
   - The naive-pooling tabular method uses RLSVI by merging the historical and target data.
   - The clustering method adaptively pools the target data and groups of historical data according to the distance of the estimated transition probability for each (h,a) pair; see Miao et al. (2022) for motivation of this method and Appendix EC.4.5 for adaptation to our RL setting.
2. Contextual-based methods:
   - The `no-pooling contextual` method uses RLSVI with linear Q-function approximation and without pooling.
   - The `naive-pooling contextual` method fits a regression model for the Q-function by merging the historical and target data.

The four benchmark algorithms in the tabular category require the classification of target patients as we did for `adaptive-pooling tabular`. The other two benchmarks are contextual-based methods so do not require classification; they can be seen as the contextual version of the no-pooling and naive-pooling methods, respectively. The implementation details of these algorithms are in Appendix EC.4.5. Moreover, to give enough "benefit of the doubt" to the benchmark algorithms, we tune the necessary hyper-parameters for each benchmark and choose the best one to report. All the hyper-parameters are specified in Appendix EC.4.6.

7.4. Evaluation Results

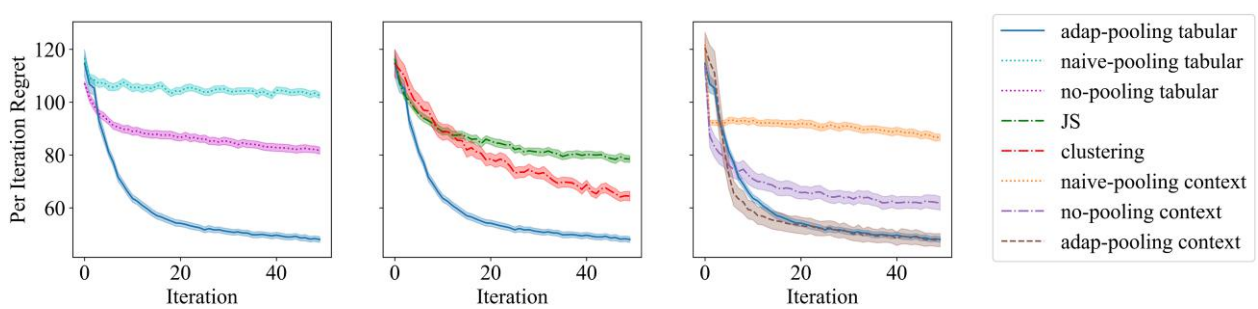
7.4.1. Baseline Results. Figure 1 displays the regret performance of our data-pooling algorithms and other benchmarks. We can see that the average regret from either of our methods (the blue and brown curves) decreases much faster than others, especially in the first 10 iterations—this is where the small-sample issue is particularly prominent. Table 2 reports the numeric values of the total regret for each algorithm, as well as the total cost, the main patient outcome (30-day readmission rate), and the computational time. From the table, it is clear that our data-pooling algorithms achieve the smallest total regret, the smallest cost, and the lowest readmission rate over 50 iterations. Comparing the two adaptive-pooling methods, the one that incorporate the contextual learning among target patients slightly improves over the tabular version with faster computation, which is expected.

7.4.2. Explanation. The `no-pooling tabular` method, which uses only target data and learns each class separately, struggles with limited samples. This results in significant regret during the initial iterations and slower convergence (Figure 1). The `no-pooling contextual` method performs better due to cross-learning among target patients but still suffers from small sample issues in the early stages. In contrast, the `naive-pooling tabular` method (light blue curve) is overly influenced by historical data, which differs from target patients (Table 1). Its reliance on historical sample sizes slows policy updates, leading to continued suboptimal results and high regret after 50 iterations. Similarly, the `naive-pooling contextual` method (orange curve) achieves lower regret than its tabular counterpart but remains limited by its regression approach, which fails to capture population heterogeneity via the assumed parametric model. The `JS` method focuses on estimation accuracy without optimizing decision-making quality when pooling the target and historical data, which results in subpar performance. The `clustering` method (red curve) performs reasonably well, with a 34% performance gap compared with our adaptive-pooling methods. The clustering method groups target patients with similar historical patients based on readmission risk. But unlike our "soft" grouping approach, clustering uses "hard" group inclusion/exclusion, making it sensitive to sampling variances and less stable.

7.4.3. Historical Sample Sizes. We note that although the overall sample size for the historical data are large, certain state-action pairs have relatively small sample sizes, particularly for groups with lower intervention rates. The results shown in Figure 1 and Table 2 indicate the robustness of our algorithm to uneven historical sample sizes. We have also conducted additional analyses by randomly removing historical data tuples and the performance of our data-pooling algorithm remained superior; see Appendix EC.4.7.2 for more details.

7.4.4. Sensitivity Analyses in Synthetic Setting. We evaluate the robustness of our algorithm through extensive sensitivity analyses. Specifically, we utilize a synthetic setting designed to closely resemble the baseline (real-data calibrated) scenario, which allows for a more comprehensive evaluation of the algorithm's performance across various conditions. In the synthetic

Figure 1. (Color online) Regret per Iteration of 146 Target Patient Classes



Notes. The solid line represents the average from 100 replications, and the shaded area corresponds to the 95% confidence interval. We separate the comparison into three plots to avoid lines overlapping with each other; adaptive pooling tabular is represented by the blue line and is shown in all three plots.

Table 2. Performance of Different Algorithms in 50 Iterations Estimated via 100 Rounds of Simulation

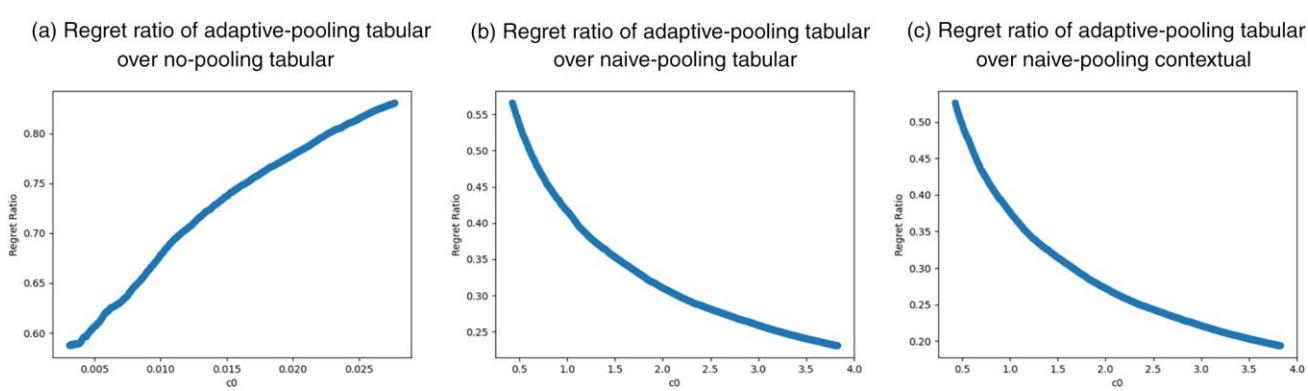
Algorithm	Total regret	Total cost	Readmission rate	CPU time/class
adaptive-pooling tab.	2,401.71 ± 54.44	50,251.76 ± 54.44	9.83%	24.92
adaptive-pooling contx.	2,390.48 ± 124.59	50,240.53 ± 124.59	9.92%	13.26
JS	3,923.05 ± 63.33	51,773.10 ± 63.33	10.29%	17.01
no-pooling tab.	4,401.91 ± 60.91	52,251.96 ± 60.91	10.26%	7.67
naive-pooling tab.	5,123.83 ± 54.43	52,973.88 ± 54.43	10.26%	6.89
clustering	3,220.99 ± 89.24	51,951.78 ± 89.24	10.34%	14.95
no-pooling contx.	2,933.65 ± 111.01	50,783.70 ± 111.01	10.75%	4.87
naive-pooling contx.	4,332.42 ± 62.82	52,182.47 ± 62.82	10.82%	3.84

Notes. The numbers following the ± sign are half-width of the 95% confidence intervals. CPU times are measured in second.

setting, we vary the following for sensitivity analyses: (i) the state and action spaces, (ii) constraints on the number of switches, (iii) the gap between the historical and target patients, Δ, and (iv) the number of target learning iterations, T. In particular, the state space becomes the three-dimensional $s = (s_1, s_2, s_3)$ and the action space is $\mathcal{A} = \{0, 1, 2\}$ with two types of interventions as described in Section 7.1. Table EC.11 in Appendix EC.4.7 summarizes all the sensitivity check we have performed. Through performance comparison with two tabular benchmarks (`no-pooling tabular` and `naive-pooling tabular`) and a contextual benchmark (`naive-pooling contextual`), we convey two main messages: (1) our algorithm consistently outperforms the benchmarks across various settings, and (2) we demonstrate when the bias in historical data leads to greater performance improvement, substantiating the asymptotic results from Theorem 2 and highlighting the importance of a well-designed pooling mechanism – since naive-pooling can sometimes perform worse than not using historical data at all. Below, we provide the key findings related to message (2), while the details of the experimental setup and algorithm performance comparisons are provided in Appendices EC.4.8.

7.4.5. Importance of Proper Pooling. Figure 2 shows the regret ratio when we change Δ and T. Figure 2(a) shows a setting with a small Δ, less target samples, and focuses on the $c_0 < 1$ region (corresponds to the first part of the results in Theorem 2). Figure 2, (b) and (c) are from a setting with a large Δ, more target samples, and focus on the $c_0 > 1$ region (corresponds to the second part of Theorem 2). We can see the trends in all three plots are consistent with the theoretical insight provided by Theorem 2. That is, in plot (a), the regret ratio between our method and no-pooling decreases as c_0 goes to 0 (from decreasing T), which indicates the performance gap between the no-pooling and ours enlarges. This is expected since more reliance should be put on historical data when the target data are limited. In contrast, plots (b) and (c) show that the regret ratio between our method and naive-pooling decreases as c_0 increases. This corresponds to the regime where we should put more reliance on target data and less on historical data. Naive-pooling's performance deteriorates significantly since it does not properly account for the bias caused by the population difference, even as the target data accumulates; a similar explanation applies to the contextual version. In fact, the naive-pooling's performance can get arbitrarily bad when increasing Δ – it becomes even worse than not using any historical data at all. This reiterates the importance of a proper pooling mechanism. More numerical results are in Appendix EC.4.8.3.

Figure 2. (Color online) The Regret Gap Between Adaptive-Pooling vs. Other Benchmarks



Note. Plot (a) and plots (b) and (c) use two different parameter regimes, and the detailed numerical settings are in Appendix EC.4.8.3.

7.5. Practical Implications

Improving decision-making quality in data-limited environments is crucial for healthcare management and other domains requiring tailored interventions. When a problem can be well-approximated without model misspecification (e.g., individuals from the same population), contextual-based parametric approaches often demonstrate superior performance in practical applications. However, the scenario becomes more intricate when the possibility of model misspecification enters the equation. In such cases, caution should be taken when using the conventional wisdom of pooling data. Our study shows that leveraging historical data but without considering the underlying differences among populations could result in even worse performance than not using historical data at all. The superior performance our data-pooling algorithm highlights the critical importance of designing the right pooling mechanism to balance variance reduction from larger sample sizes with potential bias from population differences.

Moreover, the success of data-sharing via aggregate statistics shown in this paper provides a viable option for individual organizations to leverage public datasets or published studies to augment limited in-house data. This has important implication for enabling health organizations to adapt quickly to new environments or patient populations, such as identifying effective interventions when transitioning between regions with different treatment histories. It also offers a pathway for public policy-makers to encourage organizations to share their aggregate data to facilitate population health outcomes for the broader community.

While our primary motivation and application context are in healthcare, the growing needs of personalized intervention are pervasive in many applications, as are the challenges associated with the small-sample issue. Our development can be used to support similar decision-making problems, such as incarceration diversion decisions for reducing recidivism in criminal justice systems (Zhang et al. 2022); revenue management (deciding which customers to follow up with to increase their chance of subscribing to paid services); and pricing (Bastani et al. 2022).

8. Conclusion and Future Work

In this paper, we show that managing preventative intervention planning for a panel of heterogeneous patients is nontrivial and cannot rely solely on heuristics or human experience. To tackle the small-sample issue in personalization and improve performance over existing learning algorithms, we propose a novel data-pooling algorithm that properly incorporates data collected from other individuals (e.g., historical data, published studies), overcoming shortages associated with existing methods. We establish theoretical performance guarantees for our data-pooling algorithm and the improvement in regret bound over other benchmarks. We also demonstrate its empirical success via a case study using real data and synthetic experiments, where extensive simulation results indicate our algorithm's potential to reduce readmissions in a post-discharge setting.

Regarding future directions, on the theoretical side, the current RL formulation we consider requires interactive engagement with target patients, that is, on-policy evaluation. We leave the extension to the off-policy setting as an important but challenging future research direction. Future research could also investigate more general conditions under which similar asymptotic results could be obtained and provide further theoretical justification for the design of data-pooling contextual estimators where data from all target patients are used in real time. On the practical side, while our simulation results are encouraging, additional real-world testing is essential to fully validate its effectiveness and practical applicability in diverse healthcare settings. We are in the process of working with some partner hospitals to design randomized control trials in the readmission context to evaluate the algorithm's performance in a real environment. Moreover, although this paper focuses on noninvasive interventions in preventative healthcare, our approach effectively utilizes historical data as a "warm start." Rather than starting from scratch, we leverage prior knowledge to guide recommendations in new environments. Any deviations from established procedures are more about refining than radical or potentially harmful changes. This approach promotes safe exploration, a principle relevant to broader clinical contexts, such as updating hospital care protocols. The framework developed in this paper lays a promising foundation, and an important avenue for future work involves further ensuring safe exploration.

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Xinyun Chen and Pengyi Shi contributed equally.

Endnotes

[1] Given our focus on preventative interventions, which typically involve non-invasive actions related to administrative or informational processes and do not directly impact a patient's physiological well-being, explorations inherent in learning do not pose immediate physical risks to patients. Hence, the online learning framework is a suitable tool.

[2] The cost is not normalized to [0, 1], as in the theoretical analysis, but that does not affect our comparison. The values for $c(a)$ and c_R are set to ensure enough difference in the optimal policies across patients. The values can be varied to generate tradeoff curves—tradeoff between the number of interventions versus the readmission rates.

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